MIL-T-5483D(WP)
1 December 1964
SUPERSEDING
MIL-T-5483C(Wep)
1 July 1960

MILITARY SPECIFICATION

TRANSMITTERS, PRESSURE, SYNCHRO, AIRCRAFT, 165-DEGREE MOVEMENT

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers hermetically sealed 165-degree movement synchro type aircraft pressure transmitters.

1.2 Classification - The transmitters shall be furnished in the following ranges:

MS PART NO.	RANGE	
MS28130-1	0-20 PSI	Differential Pressure
MS28130-2	0-20 PSI	Gage Pressure
MS28130-3	0-50 PSI	Differential Pressure
MS28130-4	0-50 PSI	Gage Pressure
MS28130-5	0-100 PSI	Differential Pressure
MS28130-6	0-100 PSI	Gage Pressure
MS28130-7	0-200 PSI	Differential Pressure
MS28130-8	0-200 PSI	Gage Pressure
MS28130-9	0-300 PSI	Differential Pressure
MS28130-10	0-300 PSI	Gage Pressure
MS28130-11	0-500 PSI	Differential Pressure
MS28130-12	0-500 PSI	Gage Pressure
MS28130-13	0-1000 PSI	Gage Pressure
MS28130-14	0-1000 PSI	Approximate Gage Pressure
MS28130-15	0-2000 PSI	Approximate Gage Pressure
MS28130-16	0-5000 PSI	Approximate Gage Pressure

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

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SPECIFICATIONS

Federal

BB-N-411	Nitrogen

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-I-5482	Indicators, Pressure, Synchro, Aircraft, 165-Degree Scale
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-756	Reliability of Weapons Systems, Procedure for Prediction and Reporting Prediction of

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Military (Continued)

MS28130	Transmitters, Pressure, Synchro, Aircraft, 165-Degree Movement
MS33586	Metals, Definition of Dissimilar
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

2.2. Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Qualification - The transmitters furnished under this specification shall be a product which has been tested and meets the Quality Assurance Provisions specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of Government documents - Except as provided in 3.2.1 and 3.2.2, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, when on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, and cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

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3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected, as specified herein, to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the transmitter may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals, as defined in MS33586, shall not be used in intimate contact with each other, unless protection against electrolytic corrosion is provided.

3.3.4 Protective treatment - When materials are used in the construction of the transmitter that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If the materials are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors - Materials used in the construction of the transmitter shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - The transmitter shall be designed for use with pressure indicators conforming to Specification MIL-I-5482. The rotor of the transmitter synchro shall rotate, relative to the stator, through an angle of 165 ±1 degrees for the range of pressures specified in paragraph 1.2. The transmitter shall be provided with means to adjust or correct the indications. The means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment can be accomplished from the exterior of, and without damage to, the hermetically sealed case. The adjustment shall provide a minimum of 14 degrees of indicator adjustment. The transmitting mechanism shall be distinct from the case so that small distortions of the case will not affect the accuracy of the indications. Each transmitter shall be capable of operating either one or two indicators without change in indication of more than one degree of arc. The transmitter shall be

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constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.4.1 Maintenance - The design shall be such as to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment, which are normally available as commercial standards.

3.4.2 Case - The outline dimensions and markings of the transmitter shall conform to the requirements of MS28130. The case shall be finished in lusterless black material, Color No. 37038 of FED-STD-595.

3.4.2.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed, replaced, and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any environment to which the transmitters may be subjected in normal use in military aircraft.

3.4.2.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with Specification BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ±0.1 atmosphere.

3.4.2.3 Case pressure release - A metal blow-out device to discharge excessive case pressure shall be provided in each pressure transmitter in the ranges of 0-500 PSI through 0-5,000 PSI. This device shall operate to release the internal case pressure through an approximately 0.045-inch-diameter orifice whenever the internal case pressure, relative to the atmosphere, reaches 90 ±30 PSI. The device shall not affect hermetic sealing. The design of the blow-out device shall be submitted to the procuring activity for prior approval.

3.4.3 Differential pressure transmitters - Each differential pressure transmitter shall have two pressure connections, and they shall be marked as shown on MS28130. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressures applied to the two pressure connections; and the angular position of the rotor of the synchro relative to its stator shall be practically a linear function of $(P_1 - P_2)$.

3.4.3.1 When a case pressure release is required, the blow-out device shall discharge through a 5/16-24 connection, conforming to MS33656, in an optional location on the back plate of the case. The case pressure release connection shall be marked "CPR". The transmitter shall be so designed and constructed that if either the "P_1" pressure sensor

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or the "P_2" pressure sensor leaks, the pressure in the case will be relieved, and the escaping fluid discharged through the case pressure release connection.

3.4.4 Gage pressure transmitters - Each gage pressure transmitter shall have two pressure connections and they shall be marked as shown on MS28130. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the pressure ("P") port and the pressure applied to the vent ("V") port, and the angular position of the rotor of the synchro relative to the stator shall be practically a linear function of (Pressure $_P$ - Pressure $_V$).

3.4.4.1 The 500 and 1000 psi transmitters shall have a case pressure release constructed so that if the sensor connected to the "P" port should leak, the case pressure will be relieved and the fluid discharged through the Vent port. This port shall be marked "V & CPR".

3.4.5 Approximate gage pressure transmitters - Each approximate gage pressure transmitter shall have one pressure connection. The pressure connection shall be marked "P". The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the pressure connection and the pressure in the case of the transmitter, and the angular position of the rotor of the synchro relative to the stator shall be practically a linear function of (Pressure $_P$ - Pressure $_{Case}$).

3.4.5.1 The blow-out device for releasing case pressure shall discharge through a 5/16-24 connection conforming to MS33656. The transmitter shall be so designed and constructed that if the pressure sensor leaks, the pressure in the case will be relieved, and the escaping fluid discharged through the case pressure release connection.

3.4.6 Stops -

3.4.6.1 Overpressure stop - All transmitters - Each transmitter shall include an overpressure stop which shall act directly on the pressure sensor(s) and which shall limit the overtravel of the rotor of the synchro beyond its full scale position, to some value between one degree and fifteen degrees inclusive.

3.4.6.2 Reverse pressure stop - Each gage pressure transmitter, up to and including 50 psi, and all differential pressure transmitters shall include a device which will limit the reverse rotation of the rotor of the synchro beyond its zero pressure position to some value between one degree and thirty degrees, inclusive.

3.4.7 Weight - The weight of the completely assembled transmitter shall not exceed 1.4 pounds.

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3.4.8 Screw threads - Screw threads, 0.060 inch in diameter or larger, shall be in accordance with Specification MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Primary input power requirements - The transmitter shall operate on AC power (single phase), 26V ±2 volts, 400 cps ±5 percent, 6.24 VA maximum.

3.5.2 Undervoltage protection - The transmitter shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

3.5.3 Electronic parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity.

3.5.4 Electric connectors - The electric connector used shall conform to Specification MIL-C-5015 and MS28130.

3.5.5 Wiring - The internal and external wiring shall be as shown in Figure 1 and Figure 2, respectively. Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. The wiring shall be insulated from the transmitter case. The signal rotation at the pins of the electrical connector of the transmitter shall be ADC for increasing value of the pressure function being transmitted.

3.6 Performance - The transmitter shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.6.1 Reliability - The transmitter reliability shall be analyzed, calculated, and reported in accordance with MIL-STD-756.

3.6.2 Reliability in mean-time-between failures - The mean-time-between failures (MTBF) shall be measured in cycles as defined in 4.6.18. The transmitter shall have a minimum of 120,000 operating mean cycles between failures, when tested and accepted as outlined under the requirements of 4.4.3.

3.7 Identification of product -

3.7.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN shall be omitted.

3.7.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers, if the drawing is tabulated and covers more than one part.

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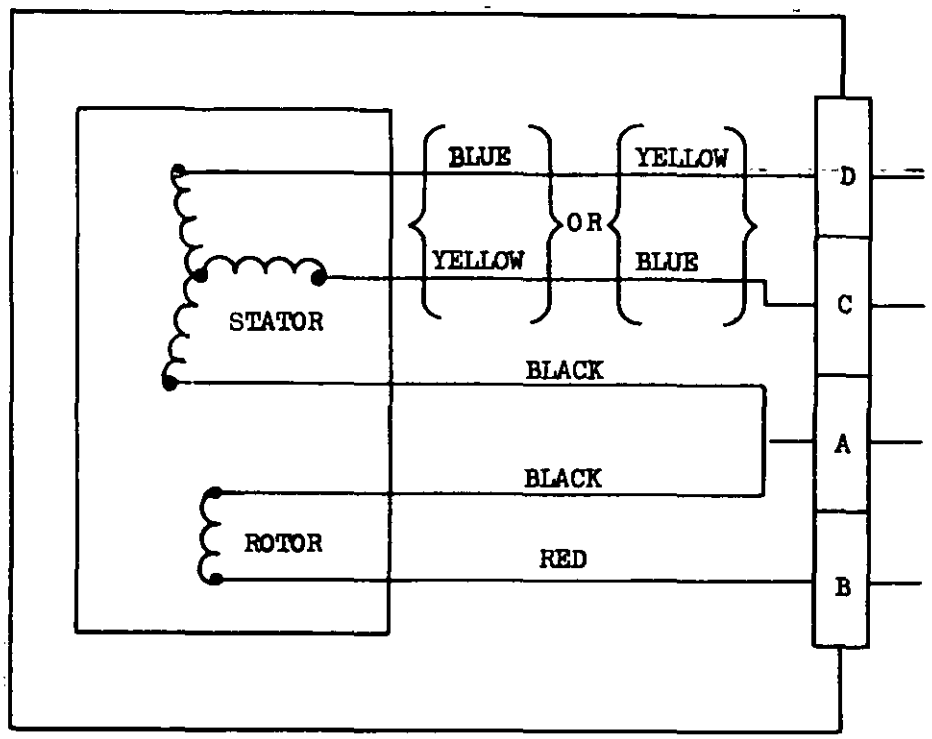
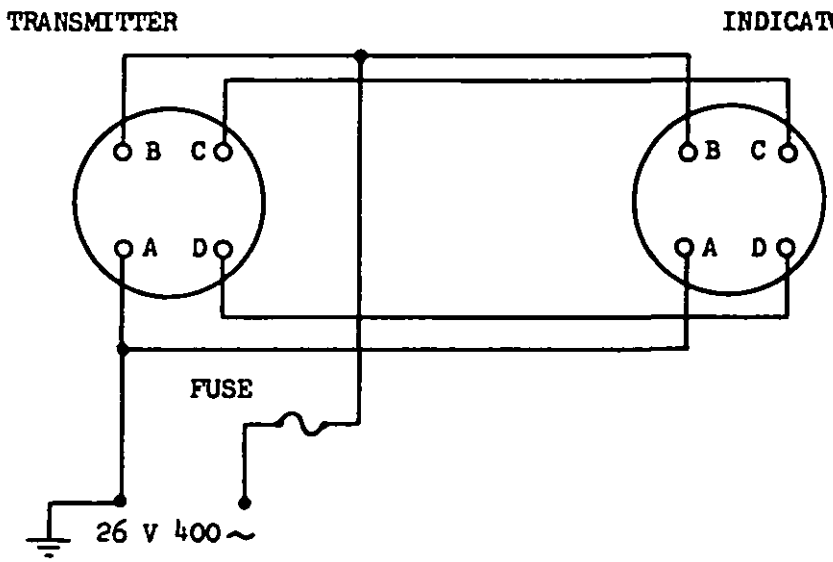
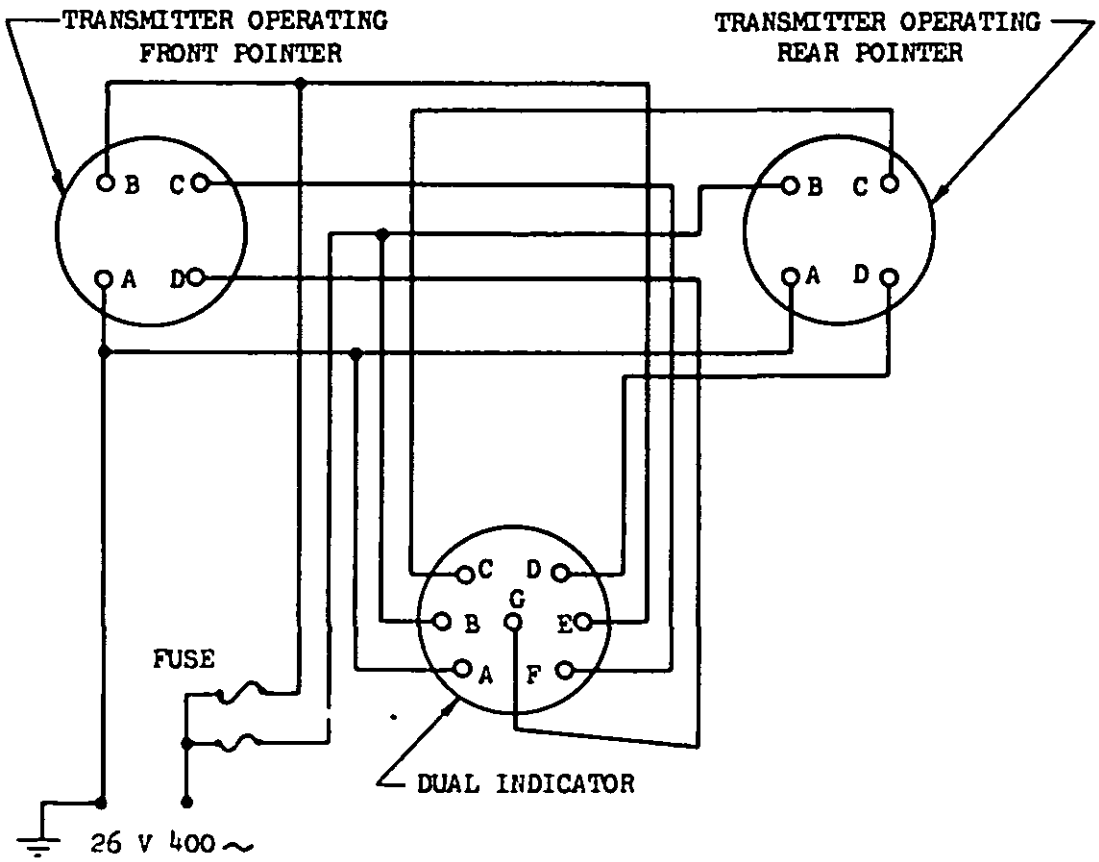


Figure 1 - Transmitter Internal Wiring Diagram

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FOR SINGLE INDICATOR



FOR DUAL INDICATOR

Figure 2 - External Wiring Diagrams

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3.8 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.9 Workmanship - The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.10 No data is required by this specification (other than reports accompanying samples submitted for qualification testing), or by applicable documents referenced in Section 2, unless specified in the contract or order. (See paragraph 6.2.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspection of the transmitter shall be classified as follows:

(a) Qualification inspection: Qualification inspection consists of examinations and tests performed on sample transmitters submitted for approval as a qualified product.

(b) Quality conformance inspection: Quality conformance inspection consists of examinations and tests performed on transmitters manufactured and submitted for acceptance under contract.

4.3 Qualification inspection - The qualification inspection of the transmitters shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection methods.

4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of three transmitters manufactured in accordance with this specification. The transmitters submitted for qualification

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inspection shall have been previously subjected only to the Individual inspections. The samples shall be forwarded at the contractors expense to the laboratory designated in the Letter of Authorization.

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification inspection
Transmitter, Pressure, Synchro, Aircraft, 165-Degree Movement
Submitted by (*Manufacturer's name, date*) for Qualification inspection in accordance with Specification MIL-T-5483 under authorization (reference letter authorizing tests).

4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual inspection, the Sampling plans, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required inspections. Sampling plan B shall be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspection, except for Sampling plan B, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the final acceptance of the finished product.

4.4.1 Individual inspection - Each transmitter submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each transmitter accepted shall have passed the following inspections:

Examination of product
Scale error at room temperature
Friction error
Position error
Dielectric strength
Sensors matching (for differential transmitters only)
Proof pressure
Sealing

4.4.2 Sampling plans - The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished (see 6.4.5) and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

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4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule.

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	See Note
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective transmitter occurs, no transmitters from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the following inspections in the order listed.

Suction
Voltage and frequency variation
Seasoning
High temperature
Low temperature
Vibration error

4.4.2.2 Sampling plan B instructions - Two transmitters shall be selected at random, from the first 15 produced on contract, and submitted within 10 days after manufacture. These samples shall be forwarded at the contractors expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached, and marked with the following information:

Transmitter, Pressure, Synchro, Aircraft, 165-Degree Movement
Submitted by (Manufacturer's name, date) for Production Acceptance Sampling Plan B inspection, in accordance with Contract/order No.
Manufacturer's part number _____

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4.4.2.2.1 Sampling plan B inspection - Each sample selected for Sampling plan B inspection shall be subjected to the following inspections in the order listed.

Sampling plan A
Vibration failure
Low temperature exposure
High temperature exposure
Endurance
Overpressure
Salt spray
Case pressure release
Case proof pressure
Test for helium
Internal examination

4.4.3 Reliability tests - Test samples which have successfully passed the Individual inspections shall be chosen for Reliability Quality Assurance tests. There shall be two kinds of tests:

(a) Predelivery tests - Predelivery tests are tests conducted prior to delivery of initial production transmitters or production samples or transmitters representative of production samples. The number of samples shall be a minimum of three selected as in the order produced. The maximum number to be used as test samples is at the discretion of the contractor.

(b) Delivery tests - Delivery tests are those tests conducted on transmitters starting with initial delivery and continuing through the balance of the contract. Samples shall be selected in the order produced. Selection shall be 1 out of each 15 produced. The contractor may elect to use additional samples, however, the number shall not be so large as to adversely affect the delivery requirements of the contract.

All reliability tests shall be conducted by the contractor. A Detailed Reliability Test Procedure Outline shall be submitted to and approved by the procuring activity prior to the initiation of the tests. The test procedure must include, but is not limited to:

(a) Duty cycle.

(b) Performance characteristics to be measured.

(c) Failure criteria.

In any event, the requirements of the following subparagraphs shall constitute minimum procedures and criteria.

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Where a contractor is in continuous production from contract to contract he may request a waiver of Predelivery tests if he has already met the requirements for identical items, or similarly identical items. The request shall be additional to, but shall accompany the above Outline, and therefore be subject to approval by the procuring activity.

4.4.3.1 Procedures - The following procedures shall be adhered to for the respective tests:

(a) Predelivery tests.

1. Failure of sample in a way repairable - The sample may be repaired, or replaced, and the tests continued. The choice is the contractor's.

2. Failure of a sample in a way not repairable - The sample must be replaced if three only were used initially.

3. New items placed on test may have design corrections incorporated. This should be noted in the log.

4. To be counted toward the MTBF determination, a total of 3 times the MTBF (as a minimum) must be run on the items.

5. Two samples must demonstrate a MTBF at least equal to the required MTBF. (Not necessarily without failure.)

6. In computing the MTBF, equipments, their equipment running time, and failures may be omitted (censored) provided at least three consecutive equipments of the latest entry are used and all design changes incorporated therein are in the production equipments.

7. All failures shall be analyzed and corrections made in production when appropriate.

8. A log shall be maintained in which pertinent information must be entered. This log shall be available to the procuring activity.

(b) Delivery tests.

1. Failures must be analyzed and production corrections incorporated as applicable.

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2. Test results must satisfactorily meet the requirements of the following table:

Failures	Total in Multiples of Specified MTBF (Operating Cycles)
0	4.4
1	5.2
2	6.0
3	6.8
4	7.6
5	8.5
6	9.3

3. All samples must count toward the final MTBF determination unless otherwise agreed to by the procuring activity.

4. Test results must be recorded, and be available to the procuring activity.

5. At least one sample shall run the full required MTBF period, without failure.

4.4.3.2 Test level - The test level shall be III from Table I of Specification MIL-R-23094.

4.4.3.3 Duty cycle - The duty cycle shall be continuous. The duty cycle shall consist of cycling the transmitter as described in the Endurance test.

4.4.3.4 Performance characteristics to be measured - The Scale error and Friction error tests, described under Inspection methods, shall be conducted at least once each day.

4.4.3.5 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.4), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

4.4.3.6 Preventive maintenance - No preventive maintenance may be accomplished on the samples while they are on test.

4.4.3.7 Disposition of samples upon completion of reliability tests - Any sample tested may be delivered on contract provided it meets all the following requirements:

(a) It is representative of production units currently being accepted.

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(b) It is in "good as new" condition or has been refurbished. (See 6.4.4 and 6.4.5)

(c) It is otherwise satisfactory.

4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all inspection required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5° C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg).
Humidity	Room ambient up to 90 percent relative humidity.
Input Power	26 ±0.5 V AC, 400 ±5 cps.

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken, the transmitter shall be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch. Also, unless otherwise specified, two sets of readings shall be taken; the pressure shall be brought up to but not exceed each test point, then the pressure shall be brought down to but not fall below each test point.

4.5.3 Attitude - Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.5.4 Test standards - The transmitter shall be tested against an indicating standard constructed of a standard production indicating unit selected for a maximum error of ±0.5 of a circular degree against a primary transmitting master. A primary transmitting master shall be a selected transmitting standard having a maximum inherent error of ±0.1 of a circular degree.

4.6 Inspection methods -

4.6.1 Examination of product - Each transmitter shall be examined to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Scale error at room temperature - The transmitter shall be tested for scale error at room temperature. The test points and tolerances shall be as specified in Table I.

4.6.3 Friction error - This test may be combined with the Scale error at room temperature test. The transmitter shall be tested for friction error at the test points specified in Table I. Two readings shall be taken; the first before and the second after the transmitter has

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TABLE I

Test Points and Tolerances for Test Conditions
Tolerances are Plus or Minus PSI

Range, PSI	0-20	50	100	200	500	1000	2000	5000
Test Points, PSI	0 4 8 12 16 20	0 10 20 30 40 50	0 20 40 60 80 100	0 40 80 120 160 200	0 100 200 300 400 500	0 200 400 600 800 1000	0 400 800 1200 1600 2000	0 1000 2000 3000 4000 5000
Overpressure, PSI	40	100	200	400	1000	1500	3000	6000
Tolerances, PSI								
Scale error at room temperature	0.4	1.0	2.0	4	10	20	40	100
Friction error	0.2	0.5	1.0	2	5	10	20	50
Position error	0.2	0.5	1.0	2	5	10	20	50
Sensors matching (for differential transmitters only)								
1/5 Range	0.5	1.2	2.5	5	12	-	-	-
2/5 Range	0.6	1.5	3.0	6	15	-	-	-
3/5 Range	0.7	1.8	3.5	7	18	-	-	-
4/5 Range	0.8	2.0	4.0	8	20	-	-	-
Full Range	0.9	2.3	4.5	9	23	-	-	-
Vacuum	0.5	1.2	2.5	5	12	-	-	-
Seasoning	0.4	1.0	2.0	4	10	20	40	100
High temperature	0.6	1.5	3.0	6	15	30	60	150
Low temperature	0.8	2.0	4.0	8	20	40	80	200
Vibration error	0.2	0.5	1.0	2	5	10	20	50
Vibration failure	0.4	1.0	2.0	4	10	20	40	100
Endurance	0.7	2.0	3.5	7	20	40	80	200
Scale error overpressure	0.4	1.0	2.0	4	10	20	40	100

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been vibrated as in 4.5.2. The difference between the first and second readings shall not exceed the tolerances specified in Table I.

4.6.4 Position error - The position error tests may be conducted in conjunction with the Scale error at room temperature test. The transmitter, mounted in its normal operating position, shall be energized to obtain a reading of approximately 1/2 scale. Additional readings shall then be taken when the transmitter is in each of the following positions:

90° up, 90° down (longitudinal axis vertical).

90° CW, 90° CCW from vertical position above, (longitudinal axis horizontal).

The maximum change in indication from that in the normal operating position shall not exceed the tolerance specified in Table I.

4.6.5 Dielectric strength - A potential of 200 volts (root mean square) alternating current, at commercial frequency, shall be applied between isolated pins, and between pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the transmitter as a result of this test.

4.6.6 Sensors matching - (Differential pressure transmitters only)

(a) A pressure, relative to the atmosphere, equal to 1/5 of the range of the transmitter, shall be applied to the "P_2" port of the transmitter. A pressure shall be applied to the "P_1" port which shall exceed that applied to the "P_2" port by each of the test points listed in Table I, except any test point which requires that a pressure be applied to the "P_1" port the value of which, relative to the atmosphere, exceeds the value listed on the "Overpressure" row of Table I.

(b) The test of (a) above shall be repeated substituting "2/5" for "1/5" wherever it appears.

(c) The test of (a) above shall be repeated substituting "3/5" for "1/5" wherever it appears.

(d) The test of (a) above shall be repeated substituting "4/5" for "1/5" wherever it appears.

(e) The "P_1" and "P_2" ports shall be connected together and one pressure shall be applied to the two ports simultaneously. The value of this pressure, relative to the atmosphere, shall be slowly varied continuously

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from zero through the value listed on the "Over-pressure" row of Table I, and back to zero. The indication shall not differ from zero at any time by more than the value called out in the "Scale Error - Sensors Matching - Full Range" row of Table I.

(f) A vacuum of 14.0 ±0.5 psi shall be applied to the "P_2" port of the transmitter. A pressure shall be applied to the "P_1" port which shall exceed that applied to the "P_2" port by each of the test points. The scale error shall not exceed the value listed in the "Scale Error-Sensors Matching-Vacuum" row of Table I.

4.6.7 Proof pressure - For differential pressure transmitters in the range of 0-500 psi and all approximate gage pressure transmitters, a pressure of 75 psi, relative to the atmosphere, shall be applied to the case pressure release connection of the transmitter for ten minutes. The sealing test shall be performed on the transmitter after the application of this pressure.

4.6.8 Sealing - Pressure and vent connections shall be closed with suitable plugs for this test. Each transmitter shall be immersed in a suitable deaerated liquid, such as water, within an enclosure. The absolute pressure of the air above the liquid shall be reduced to 2-1/2 inches Hg. Any bubbles coming from within the transmitter case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air in the various exterior parts of the case shall not be considered as a leak. A wetting agent shall be used in the liquid to facilitate removal of surface bubbles.

4.6.8.1 A leakage test by other methods, equivalent or superior in sensitivity to the above described immersion test, may be used upon approval by the procuring activity. Where a helium leak detector is employed for conducting this test, the leak rate shall not exceed 1 micron-cubic foot per hour at a pressure differential of approximately 1 atmosphere.

4.6.9 Suction - Gage pressure transmitters, with MS Part Nos. 28130-2, and -4 shall have a vacuum of 14.0 ±0.5 psi applied to the "P" port for 5 minutes with the "V" port open to atmosphere. The transmitter shall then be subjected to and shall meet the requirements of the Scale error at room temperature and Friction error tests.

4.6.10 Voltage and frequency variation - The transmitter shall be pressurized and a scale reading taken when the voltage and frequency are within the limits specified under Inspection conditions. Under the same condition of pressure, the voltage and frequency shall be adjusted as listed and respective scale readings taken.

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Percent Volts	Percent Frequency
90	95
110	105
90	105
110	95

No one of the latter four readings shall differ from the first reading by more than the tolerance given in Table I for Scale error at room temperature, for the pressure applied.

The test shall be conducted at three different pressure levels approximately equally spaced over the transmitter range.

4.6.11 Seasoning - The Seasoning test shall be conducted as is the Endurance test except that the number of cycles shall be 10,000 and the tolerances of Table I for Seasoning shall apply.

4.6.12 High temperature - The transmitter shall be placed in a chamber and subjected to a temperature of +70° ±2° C for a period of 4 hours. The transmitter shall be energized throughout the test. At the end of this period and with the transmitter still at this temperature, the transmitter shall be subjected to a scale error and friction test. The scale errors shall not exceed the tolerances specified in Table I for high temperature. The friction errors shall not exceed the tolerances specified in Table I for Friction error. Upon return to room temperature, the transmitter shall be subjected to and shall meet the requirements of the Scale error at room temperature and Friction error tests.

4.6.13 Low temperature - The transmitter shall be placed in a chamber and subjected to a temperature of -54° ±2° C for a period of 4 hours. At the end of this period and while still at that temperature, the transmitter shall be subjected to Scale error and Friction error tests. The scale errors shall not exceed the tolerances specified in Table I for low temperature. The friction errors shall not exceed the tolerances specified in Table I for Friction error. Upon return to room temperature, the transmitter shall be subjected to and shall meet the requirements of the Scale error at room temperature and Friction error tests.

4.6.14 Vibration error - The transmitter shall be connected to an indicator, energized, and a pressure applied equal to approximately midscale. The transmitter shall then be vibrated in accordance with Procedure IV of Specification MIL-E-5272. No condition of applied vibration shall cause vibration errors involving pointer oscillation or variation in excess of the tolerance given in Table I for Vibration error.

4.6.15 Vibration failure - The transmitter shall be connected to an indicator, energized, and a pressure applied equal to approximately midscale. Vibration shall be applied in accordance with Procedure V of Specification MIL-E-5272 for a 3-hour period. If resonance was observed

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between 2000 and 3000 cpm during Vibration error, the transmitter shall be cycled through this range in lieu of 300 to 3000 cpm. At the completion of the vibration the transmitter shall be subjected to and meet the requirements of the Scale error at room temperature and Friction error tests, except that the tolerance for scale error shall be as given for Vibration failure in Table I.

4.6.16 Low temperature exposure - This test shall be conducted in accordance with the Low temperature test, Procedure II of Specification MIL-E-5272. The transmitter shall be checked for operation throughout its range at the -62° ±2° C temperature. The temperature shall then be raised to -54° ±2° C for a period of not less than 24 hours. While at the -54° ±2° C temperature, the transmitter shall be subjected to Scale error and Friction error tests. The scale errors shall not exceed the tolerances specified in Table I for Low temperature. The friction errors shall not exceed the tolerances specified in Table I for Friction error. Upon return to room temperature, the transmitter shall be subjected to and shall meet, the requirements of the Scale error at room temperature and Friction error tests.

4.6.17 High temperature exposure - The transmitter shall be placed in a chamber and subjected to a temperature of +70° ±2° C for a period of 50 hours. The transmitter shall be energized throughout the test. At the end of this period and with the transmitter still at this temperature, the transmitter shall be subjected to Scale error and Friction error tests. The scale errors shall not exceed the tolerances specified in Table I for High temperature. The friction errors shall not exceed the tolerances specified in Table I for Friction error. Upon return to room temperature, the transmitter shall be subjected to and shall meet the requirements of Scale error at room temperature and Friction error tests.

4.6.18 Endurance - Each transmitter, in the ranges of 0-20 PSI through 0-2000 PSI, shall be subjected to 100,000 applications of pressure to produce an indication from zero to 80 ±5 percent of the range of the transmitter, and back to zero at a rate of 30 ±10 cycles per minute. The 0-5000-PSI transmitter shall be subjected to 100,000 cycles of pressure from 2500 PSI to 3500 PSI at a rate of 40 ±20 cycles per minute. In each case the pressure shall be practically a displaced sinusoidal function of time. One ±0.1 hour after the pressure cycling, the transmitter shall be tested for Scale error at room temperature and Friction error. The tolerances for Endurance and Friction, Table I, shall apply.

4.6.19 Overpressure - A scale error test at room temperature shall be conducted just prior to the overpressure test and the readings recorded.

4.6.19.1 All transmitters - "P_1" and "P" ports - The pressure listed in the "Overpressure" row of Table I, relative to the atmosphere, shall be applied to the "P_1" or "P" port of the transmitter for ten minutes, with the "P_2", "V" or "V & CPR" port, if the transmitter has

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one, open to the atmosphere. One ±0.1 hour after the application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding the overpressure test by more than the value listed in the Scale Error Overpressure row of Table I.

4.6.19.2 Differential pressure transmitters - "P_2" port - The pressure listed in the "Overpressure" row of Table I, relative to the atmosphere, shall be applied to the "P_2" port of the transmitter for ten minutes, with the "P_1" port open to the atmosphere. One ±0.1 hour after the application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding the overpressure test by more than the value listed in the Scale error overpressure row of Table I.

4.6.19.3 Gage pressure transmitters - "V & CPR" port - A pressure of 75 PSI, relative to the atmosphere, shall be applied to the "V & CPR" port of the transmitter for ten minutes, with the "P" port open to the atmosphere. One ±0.1 hour after the application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding the overpressure test by more than the value listed in the Scale error overpressure row of Table I.

4.6.20 Salt spray - The transmitter shall be tested in accordance with Procedure I of Specification MIL-E-5272. At the end of the 50-hour period, the transmitter shall be examined. There shall be no corrosion. The transmitter shall then be subjected to and meet the requirements of the Sealing and Dielectric strength tests.

4.6.21 Case pressure release - The pressure blow-out device shall be tested either under simulated increasing case pressure, or by actual injection of pressure into the transmitter case. The device shall operate at a pressure of 90 ±30 PSI relative to the atmosphere.

4.6.22 Case proof pressure - Following the case pressure release test, the inside of the transmitter case shall be pressurized at 135 psi (gage) for 5 minutes. Any leaks in, or distortion of, the case shall be cause for rejection.

4.6.23 Test for helium - The transmitter case shall be punctured, or the filling tube cut, and the transmitter subjected to the test for helium by means of a mass spectrometer-type helium detector. Failure to detect helium shall be cause for rejection.

4.6.24 Internal examination - The case of the transmitter shall be opened and the internal mechanism examined. Any internal deterioration or damage, which could in any manner prevent the transmitter from meeting operation or maintenance requirements during service life, shall be cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Packaging - The transmitter shall be packaged in accordance with Specification MIL-P-7936, Level A or C as specified in contract or order (see 6.2). For level A packaging, the method of preservation shall be in accordance with Specification MIL-P-116, Method 1A-5, without preservative compound.

5.2 Packing - The transmitter shall be packed in accordance with Specification MIL-P-7936, Level A, B, or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The pressure transmitters covered by this specification are intended for use with pressure indicators, conforming to MIL-I-5482, or other suitable, remote-synchro pressure indicators for indicating aircraft pressures.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The quantity, MS part number and type number, if any, of the transmitter desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

(d) The laboratory that shall conduct tests (see 4.4, 4.4.2.2).

(e) Data requirements.

Certain data is required to implement 3.6.1, 4.4, and 4.4.3.

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether

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procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is used, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air-tight seal, made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets.

6.4.3 Back plate - The back plate is the surface of the case which is shown on MS28130 as having the pressure and vent connection bosses and the external zero reset cap.

6.4.4 Good as new - "Good as new" shall mean transmitters operated less than 10 percent of specified MTBF operation.

6.4.5 Refurbished - "Refurbished" shall mean that the transmitter has been completely overhauled with all component parts meeting current part standards; and the transmitter shall have been subjected to and met all the requirements of a new transmitter.

6.5 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington, D. C., 20360, and information pertaining to qualification of products may be obtained from that activity.

6.6 Changes from previous issue - The outside margins of this specification have been marked "#" to indicate where changes from the previous issue have been made. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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6.7 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specification shall apply to the extent specified.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-5483D(WP) Transmitters, Pressure, Synchro, Aircraft, 165-Degree Movement

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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